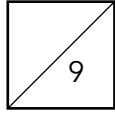


Name: _____

Time: 12 min

Class (no.): _____ ()

Score: 

Date: _____

1. Make t the subject of the formula $s = \frac{2t}{5t+1}$.

(3 marks)

$$\begin{aligned}
 s &= \frac{2t}{5t+1} \\
 s(5t+1) &= 2t \\
 5st + s &= 2t \\
 s &= 2t - 5st \\
 s &= t(2 - 5s) \\
 t &= \frac{s}{2-5s}
 \end{aligned}$$

2. Solve $\begin{cases} x+2y=11 \\ 2x-3y=-6 \end{cases}$.

(3 marks)

$$\begin{cases} x+2y=11 & \text{--- ①} \\ 2x-3y=-6 & \text{--- ②} \end{cases}$$

$$2 \times \text{①}: 2x + 4y = 22 \quad \text{--- ③}$$

$$\begin{aligned}
 \text{③} - \text{②} \quad 7y &= 28 \\
 y &= 4
 \end{aligned}$$

$$\begin{aligned}
 \text{Sub. } y=4 \text{ into ①} \\
 x + 2(4) &= 11 \\
 x &= 3
 \end{aligned}$$

$$\star \therefore x=3, y=4 \quad \text{or} \quad \begin{cases} x=3 \\ y=4 \end{cases}$$

3. There is a total of 30 \$10 coins and \$1 coins in a box. If the total value of these coins is \$120, find the numbers of \$10 and \$1 coins respectively in the box.

(3 marks)

Let the no. of \$1-coins be x and that of \$10-coins be y .

$$\begin{cases} x+y=30 & \text{--- ①} \\ 10y+x=120 & \text{--- ②} \end{cases}$$

$$\begin{aligned}
 \text{②} - \text{①}: 9y &= 90 \\
 y &= 10
 \end{aligned}$$

$$\begin{aligned}
 \text{Sub. } y=10 \text{ into ①} \\
 x + 10 &= 30 \\
 x &= 20
 \end{aligned}$$

$\therefore$ There are 20 \$1-coins and 10 \$10-coins